

TECHNICAL APPROACH

Offline-first intelligence pipeline for secure, time-critical wildlife-crime reporting

1. REPORT — React Native Mobile App

Photo/Video • GPS Location • Timestamp • Incident Details

→ Structured, evidence-based reporting

2. PROTECT — Secure Source & Evidence

Pseudonymous Tipster ID • Encryption • JWT • Role-Based Access Control

→ Protect sensitive sources and control access

3. TRANSMIT — Offline-First Synchronization

SQLite Local Storage → Secure Sync when connectivity returns

→ Reports remain available even without network connectivity

4. INTELLIGENCE — Central Control Room

NestJS API → PostgreSQL + PostGIS → Evidence Storage

→ GIS Visualization • Verification • Prioritization • Incident Correlation

5. COORDINATE & RESPOND

Authorized Officer Routing → Multi-Incident Management → Response Tracking

SYSTEM ARCHITECTURE

React Native App → SQLite + Offline Sync → Secure API → NestJS Backend → PostgreSQL + PostGIS + Evidence Storage → React Control Room → Authorized Field Officers

TECH STACK

Mobile: React Native • TypeScript • SQLite • GPS

Web: React.js • TypeScript • GIS Dashboard

Backend: Node.js • NestJS • REST APIs • WebSockets

Data: PostgreSQL • PostGIS • S3-Compatible Storage

Security: JWT • RBAC • Encryption • Pseudonymous Identity

Intelligence: Python • Incident Correlation • Explainable Analytics (*Phase 3*)

Deployment: Docker • Cloud Infrastructure

WHY THIS ARCHITECTURE?

Offline-First → Works in poor-connectivity areas

Source Protection → Minimizes unnecessary identity exposure

GIS + Prioritization → Helps officials identify actionable incidents

Centralized Workflow → Connects reporting, verification and response

Scalable Design → Supports multiple incidents, zones and organizations

INFORMATION → INTELLIGENCE → COORDINATION → ACTION